

TOMÁS MARTIN CAPORASO

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Availability: Part-Time / Asynchronous | Open to Remote (US Time Zones)

PROFESSIONAL SUMMARY

Backend-focused Computer Engineering student with hands-on experience designing, deploying, and administering self-hosted server infrastructure. Strong foundation in algorithms, data structures, and discrete mathematics, applied to real-world backend architecture across multiple independent production-scale systems spanning Go, Kotlin, and Python. Proven ability to work autonomously, manage full-stack infrastructure (Docker, Linux, networking), and lead technical initiatives within a cloud computing student organization. Fluent in English with proven capability to engage in technical and software architecture discussions. Seeking part-time or asynchronous opportunities as a Backend Developer or AI Data Trainer / RLHF Specialist, leveraging strong analytical reasoning, systems design judgment, and technical communication skills.

EDUCATION

Universidad Austral - Pilar, Argentina

Bachelor of Engineering in Computer Science (Ingeniería Informática) - In Progress, 3rd Year | Expected Graduation: 2028

Relevant Coursework: Analysis and Design of Algorithms, Operating Systems, Databases, Discrete Mathematics, Numerical Calculus

TECHNICAL SKILLS

Programming Languages: Java, Python, Go, Kotlin, SQL, Assembler

Backend & Frameworks: Spring Boot, REST APIs, JWT Authentication, OAuth (Google)

Infrastructure & DevOps: Linux (terminal-based environments, Hyprland window manager), Docker, Portainer, Tailscale, Self-Hosted Server Administration

Architecture & Data: Clean Architecture, Ports-and-Adapters (Hexagonal), Database Design, PostgreSQL, JetBrains Exposed ORM, Data Source Mapping, Container Deployment

Messaging & Protocols: MQTT, Redis (async messaging), WebSocket, Nuxt

AI / ML Integration: LLM API Integration (AWS Bedrock, Google Gemini, Ollama local inference), AWS Rekognition, YOLOv8 (Ultralytics), Vector Databases (Qdrant)

Core CS Foundations: Data Structures, Algorithm Design and Analysis, Discrete Mathematics, Operating Systems Concepts

Languages: Spanish (Native), English (Advanced/Fluent - technical and architecture discussions)

LEADERSHIP EXPERIENCE

Tech Lead, AWS Student Builder Group

Universidad Austral - Pilar, Argentina | March 2026 – Present

- Designed technical curriculum and event content for a university-wide cloud computing student organization affiliated with AWS.
- Organized and directed the group's inaugural events, structuring the technical agenda from concept to execution.
- Guided and mentored fellow students in core AWS cloud ecosystem concepts, translating complex infrastructure topics into accessible learning content.
- Coordinated cross-functional planning between student leadership and technical content delivery.

PROJECTS

VecFin - Multi-Broker Financial Aggregation Platform (Go)

Independent Project | github.com/Masega360/VecFin

- Built a production-scale backend (Go) integrating real market-data and brokerage APIs: Binance, Kraken, IOL (Argentine broker), MercadoPago, and Yahoo Finance.
- Implemented an AI-powered financial chat assistant with dual-provider integration (AWS Bedrock and Google Gemini) with automatic fallback logic, enabling natural-language portfolio queries and recommendations.
- Designed and built a social layer (communities, posts, comments, follow system) alongside a background price-alert worker and an in-app notification system.
- Built fiscal reporting with PDF generation, consolidated dashboards, and financial planning features, backed by a PostgreSQL data layer with dedicated repository implementations per domain entity.
- Implemented JWT-based authentication with Google OAuth and secured API routes via custom middleware.

KorusLedger - Event-Driven Personal Finance System (Kotlin/Spring Boot)

Independent Project | github.com/Masega360/KorusLedger

- Designed and implemented a personal finance backend using Clean Architecture, with strict separation between API, application, domain, and infrastructure layers.
- Built RESTful APIs for user, wallet, and transaction management, including real-time balance calculation and filterable transaction queries.
- Implemented the persistence layer with PostgreSQL and JetBrains Exposed ORM following the ports-and-adapters (hexagonal) pattern; wrote unit tests using in-memory fake repositories.
- Containerized the system with Docker and Docker Compose. Currently migrating the service to Go to unify the backend stack and support planned event-driven messaging (RabbitMQ) and enforced JWT authentication.

The Stalker - Real-Time Person Analytics System (IoT / Backend)

Independent Project | github.com/Masega360/the-stalker

- Designed and built a real-time person-detection pipeline integrating ESP32 camera devices, MQTT messaging (Mosquitto broker), a Python backend worker, and a Nuxt-based frontend/backend layer.
- Implemented dual computer-vision model support in the worker - AWS Rekognition (cloud inference) and YOLOv8 (local/edge inference via Ultralytics) - with a shared result-handling abstraction.
- Built a device registration and control system enabling bidirectional MQTT communication, including secure relay actuation via an authenticated HTTP API.
- Implemented authentication (login/register) and stats/zones/rules APIs on the Nuxt backend layer.

Anima - Distributed Cognitive Multi-Agent System

Independent Project | Private repository (available upon request)

- Designed a distributed multi-agent architecture (Python 3.12, asyncio) with eleven specialized modules communicating via a Redis-based async message bus, each with an explicit input/output contract.
- Built a three-tier memory system (episodic, semantic, procedural) backed by Qdrant vector storage, with an automatic temperature-based consolidation mechanism (logarithmic reinforcement, exponential decay) that promotes frequently accessed memories into abstracted knowledge.
- Implemented a risk-classification and action-approval layer (SAFE / REVERSIBLE / IRREVERSIBLE / FORBIDDEN) that evaluates every proposed system action against a fixed set of invariant safety principles before execution - directly relevant to AI safety and RLHF-style oversight design.
- Exposed system state and reasoning flow via a FastAPI REST API and WebSocket stream for real-time observability across all agents.
- Integrated local LLM inference (Ollama: Llama 3.1 8B, Qwen 2.5 7B) and embedding generation (nomic-embed-text) into the reasoning and memory pipelines.
- Orchestrated the full multi-service system with Docker Compose.

Self-Hosted Infrastructure Administration

Independent Project

- Configured and maintain a self-hosted server environment using Docker and Portainer for container orchestration and management.
- Implemented secure remote access and networking using Tailscale across distributed environments.
- Operate primarily within Linux terminal-based workflows, including customized window manager environments (Hyprrland), demonstrating deep comfort with command-line systems administration.

ADDITIONAL INFORMATION

- Comfortable with asynchronous, remote-first collaboration and self-directed project ownership.
- Strong written and verbal English communication skills suitable for technical documentation, code review, and cross-team collaboration.
- Available for part-time or task-based engagements including backend development, data annotation, model evaluation, and RLHF-related work.